Pet Translator

Bridging the Gap Between Pets and Humans

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Machine Learning (ML01)

Architecture

An End-to-End Hybrid Pipeline

- **Frontend:** Real-time audio capture via React.
- **Edge Inference:** Local, ultra-fast classification on the smartphone via **TensorFlow.js**.
- **Cloud Backend:** FastAPI server connecting to a semantic database.
- **Generative AI: Llama 3.2** generating contextual human sentences.

Audio Signal Processing

Mohamed MELLOUK

- **Datasets:** ESC-50, AudioSet, ShivaRao (Dogs), CatMeows (Cats).
- **The Challenge:** Dog sounds are *acoustic types* (bark). Cat sounds are *situational* (food).
- **Noise Reduction:** Applying `noisereduce` to clean real-world ambient sounds.
- **Isolation:** Using Voice Activity Detection (WebRTCVAD) to isolate vocalization.

Feature Extraction

Visualizing Sound

- Audio waves are converted into visual representations (**Mel-spectrograms**).
- **Standardization:** Fixed durations of 4 seconds for dogs and 2 seconds for cats.
- **Perception:** Logarithmic scaling applied on Mel frequencies to mimic biological hearing.
- **Result:** 96x96 feature matrices, formatted for Deep Learning.

Classification Models

Anas ISARTI

- **Task:** 2 classifiers, 1 par animal. Dog: bark/growl/grunt. Cat: brushing/food/isolation.
- **Challenge:** Tiny datasets (113 dog, 440 cat clips) → need a model that generalizes + an evaluation I can trust.
- **Approach:** Frozen **MobileNetV2** (feature extractor) + small trained head. Avoids overfitting vs training from scratch.
- **Evaluation:** Group-aware cross-validation (split by individual cat) → no data leakage.
- **Deployment:** Full model exported to **TensorFlow.js**, output verified identical to Python (diff = 0).

The Data Ceiling

Analyzing Limits

- **Dog model:** Solid — CV macro-F1 ~0.82–0.85.
- **Cat model:** Good on isolation, weak on food — CV F1 ~0.30–0.37.
- **Experimentation:** 5 independent levers tested (head tuning, augmentation, classifier comparison, AST backbone, focal loss + SMOTE) — all plateaued.
- **Conclusion:** Bottleneck is the dataset size (food: only 92 clips), not model capacity.

Linguistic Brain

Abir ISLAM

- **Objective:** Give a real voice and distinct personality to the pet's raw intent.
- **Dataset:** Created a synthetic dataset mapping `[Intent + Context + Personality]` to `[Sentence]`.
- **Fine-Tuning:** Trained **Llama 3.2** using LoRA on Google Colab GPUs.
- **Deployment:** Converted weights to GGUF format for lightning-fast CPU inference.

Grounding the AI

The RAG System

- **The Problem:** LLMs can hallucinate biologically inaccurate translations.
- **The Solution:** Retrieval-Augmented Generation (RAG).
- **Implementation:**
    - Built a veterinary behavioral knowledge base.
    - Embedded using `sentence-transformers` and stored in **ChromaDB**.
    - The LLM retrieves real scientific context before generating the translation.

Frontend Integration

Amine KHALIL

- **User Interface:** A modern, responsive React application with dark/light theme, three-column layout, and chat-bubble translations.
- **Pet Selector:** Allows selecting the animal (cat/dog) - personality is assigned server-side per species (haughty_cat / excited_dog) and not exposed in the UI.
- **Real-Time Feedback:** TFJS in-browser classification computes class probabilities (bark, growl, grunt / brushing, food, isolation) and displays them alongside the LLM-translated output.

Edge Deployment & UX

Amine KHALIL

- **TF.js Deployment:** Converted Keras model to GraphModel format; fixed 4 preprocessing bugs (mel scale, filter norm, power vs magnitude, input range) to match Python exactly.
- **VAD:** Energy-based silence detection (RMS < 0.015) prevents classification on silent audio, returning "no sound detected".
- **Model Loading:** Explicit `tf.setBackend('webgl').catch(() => 'cpu')` required before model load to prevent silent fails on macOS.
- **UX Research:** 3-second minimum loading delay for perception; probability bars; mock LLM fallback ensures demo works offline.

Demonstration

(Live Video Demonstration)

Conclusion

- **End-to-End Pipeline:** Successfully integrated Edge AI (TFJS) for low-latency classification with Cloud AI (Llama 3.2 + RAG) for linguistic grounding.
- **Scientific Rigor:** Demonstrated the critical impact of dataset size ("Data Ceiling") through exhaustive evaluation and cross-validation.
- **Robust Architecture:** Solved real-world constraints like ambient noise (VAD) and LLM hallucinations (RAG).
- **Final Result:** A fully functional Proof of Concept bridging audio processing, computer vision, and NLP.

Thank you.